

Answer Key — Core Mathematics Grade 10

Sub-Strand 2.4: Trigonometry 1 • Lesson 3: Sines and Cosines of Complementary Angles: $\sin \theta = \cos(90^\circ - \theta)$

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
B	D	A	A	B	D	D

USE AFTER: OBSERVE PHASE — Observe: complementary angle pairs table (25°/65° row)

Q1. Using mathematical tables, a pair finds that $\sin 25^\circ = 0.4226$. Which of these must also be equal to 0.4226?

- A. $\cos 25^\circ$
- B. $\cos 65^\circ$**
- C. $\sin 65^\circ$
- D. $\tan 25^\circ$

Correct answer: B. 25° and 65° add up to 90°, so they are complementary and $\sin 25^\circ = \cos(90^\circ - 25^\circ) = \cos 65^\circ$. The others are different ratios or different angles.

USE AFTER: OBSERVE PHASE — Observe: class inspection of the completed master table

Q2. Which pair of angles is complementary, so that the sine of one angle equals the cosine of the other?

- A. 20° and 160°
- B. 42° and 138°
- C. 35° and 65°
- D. 42° and 48°**

Correct answer: D. Complementary angles add to 90°, and $42^\circ + 48^\circ = 90^\circ$. The other pairs add to 180° or 100°, so they are not complementary.

USE AFTER: EXPLAIN PHASE — Explain: right-angled triangle labelled from both corner A and corner B

Q3. In a right-angled triangle ABC, the right angle is at C, angle A = θ and angle B = $90^\circ - \theta$. Side a is opposite angle A, side b is opposite angle B, and side c is the hypotenuse. Which ratio equals $\sin(90^\circ - \theta)$?

- A. b/c**
- B. a/b
- C. b/a
- D. a/c

Correct answer: A. From corner B, the opposite side is b, so $\sin(90^\circ - \theta) = b/c$. This is the same ratio as $\cos \theta$, because b is adjacent to angle A.

USE AFTER: EXPLAIN PHASE — Explain: Practice set (A) Simplify: express $\cos 70^\circ$ as a sine

Q4. Which of these is equal to $\cos 70^\circ$?

- A. $\sin 20^\circ$**
- B. $\sin 110^\circ$
- C. $\sin 35^\circ$
- D. $\sin 70^\circ$

Correct answer: A. $\cos 70^\circ = \sin(90^\circ - 70^\circ) = \sin 20^\circ$. Writing $\sin 70^\circ$ ignores the change to the complementary angle.

USE AFTER: EXPLAIN PHASE — Explain: Practice set (B) Solve for θ : $\sin(2\theta + 10^\circ) = \cos(\theta + 20^\circ)$

Q5. Given that $\sin(2\theta + 10^\circ) = \cos(\theta + 20^\circ)$, what is the value of θ ?

- A. 30°
- B. 20°**
- C. 10°
- D. 40°

Correct answer: B. The angles must be complementary, so $(2\theta + 10^\circ) + (\theta + 20^\circ) = 90^\circ$. This gives $3\theta + 30^\circ = 90^\circ$, so $3\theta = 60^\circ$ and $\theta = 20^\circ$.

USE AFTER: DQB PHASE — DQB: 'I now understand that...' statements and the WHAT WE NOW KNOW column

Q6. Which statement about sine and cosine of complementary angles is correct?

- A. $\cos 65^\circ = \cos 25^\circ$
- B. $\sin 65^\circ = \cos 65^\circ$
- C. $\sin 65^\circ = \sin 25^\circ$
- D. $\sin 65^\circ = \cos 25^\circ$**

Correct answer: D. $\sin \theta = \cos(90^\circ - \theta)$, so $\sin 65^\circ = \cos 25^\circ$. The relationship swaps sine and cosine; it does not keep the same ratio for both angles.

USE AFTER: MODEL BUILDING PHASE — Model: annotating the Nakuru mast diagram with the complementary angle at the top

Q7. A mast stands on level ground. From a point 40 m from its base, the angle of elevation to the top is 50° . In the right-angled triangle formed, the angle at the top of the mast is 40° . Why can an engineer use either the 50° angle or the 40° angle to work out the mast height?

- A. Because $\sin 50^\circ = \sin 40^\circ$ in the same triangle
- B. Because the tangent of 50° and the tangent of 40° are equal
- C. Because 50° and 40° are equal angles
- D. Because the two angles are complementary, so the sine of one equals the cosine of the other and both give the same triangle relationships**

Correct answer: D. The two acute angles add up to 90° , so they are complementary. This means $\sin 50^\circ = \cos 40^\circ$ and $\cos 50^\circ = \sin 40^\circ$, so either angle describes the same triangle.